Continuous-Time Reinforcement Learning for Portfolio Optimization under Partial Information

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Abstract

We review the theoretical foundations of continuous-time reinforcement learning (RL) and their application to mean-variance portfolio selection in a regime-switching market with an unobservable regime state. Central to this review is the partially observable exploratory mean-variance (POEMV) algorithm of Wu and Li (2024), which combines the exploratory stochastic control framework of Wang *et al.* (2020), the mean-variance actor–critic architecture of Wang and Zhou (2020), and the martingale-based policy evaluation of Jia and Zhou (2022) to handle partial information via Wonham filtering. Key theoretical contributions include the Gaussian optimality of the exploratory policy in the partially observed setting, the well-posedness of a degenerate parabolic PDE system characterizing the value function, and a myopic–hedging decomposition of the optimal control that accounts for latent regime transitions. We discuss the empirical performance of POEMV on three decades of S&P 500 data, where regime-awareness yields substantial reductions in out-of-sample terminal wealth variance relative to both single-regime RL and classical plug-in approaches.

Keywords: Stochastic processes, reinforcement learning, Hamilton–Jacobi–Bellman equation, mean-variance portfolio selection, regime switching, partial observability, exploratory stochastic control.

1 Introduction

Reinforcement learning has traditionally been formulated in discrete time, where an agent observes a state, selects an action, and receives a reward at a sequence of distinct steps. Many systems of scientific and practical interest, however, evolve continuously, such as robotic locomotion, physiological regulation, autonomous navigation, and financial portfolio management, which are more naturally described by differential equations than by discrete recursions. These considerations motivate continuous-time reinforcement learning, which reframes sequential decision-making using tools from stochastic processes, control theory, and the theory of partial differential equations.

The foundational treatment of continuous-time RL is Doya (2000), who defines a continuous-time temporal difference (TD) error as the instantaneous residual of the Hamilton–Jacobi–Bellman (HJB) equation and derives both actor–critic and value-gradient policy improvement strategies directly in the continuous setting. A central open question left by this work is how to handle exploration in continuous time: In a discrete setting, a stochastic policy simply randomizes over actions at each step, but in continuous time, distributing probability over actions simultaneously changes both the drift and the volatility of the state process, so a new formalization is needed. Wang *et al.* (2020) resolve this by introducing the exploratory stochastic control framework, which extends controls from point actions to distribution-valued policies and adds an entropy regularization term to the objective to sustain exploration. Wang and Zhou (2020) then apply this framework to mean-variance portfolio selection, deriving the EMV algorithm and showing that the optimal exploratory policy is Gaussian with a tractable semi-analytical value function. The partially observable exploratory mean-variance (POEMV) algorithm of Wu and Li (2024) extends EMV to a regime-switching market where the regime is unobservable, more closely matching the financial portfolio management context by incorporating Wonham filtering to track the latent state and generalizing the value function parametrization accordingly.

The objective of this review is to examine Wu and Li (2024) as a recent theoretical and algorithmic development in continuous-time RL. The preceding works of Doya (2000), Wang *et al.* (2020), and Wang and Zhou (2020) are treated as necessary background, providing the exploratory control framework and mean-variance machinery that Wu and Li (2024) extend. Our focus is on the theoretical advances that this extension requires—particularly the well-posedness of a degenerate PDE system, the survival of Gaussian optimality under partial information, and the myopic–hedging decomposition of the optimal control—and on the empirical evidence that regime-awareness materially improves out-of-sample performance.

The remainder of this review is organized as follows. Section 2 describes the exploratory stochastic control framework and its mean-variance application, in addition to the regime-switching extension, stating the motivation and formulation of the main research questions addressed by Wu and Li (2024). Section 3 establishes the preliminary analytical results that make the problem tractable, including the degenerate PDE theory required by the partial-information structure. Section 4 details the main algorithmic and theoretical contributions of Wu and Li (2024). Section 5 reports the simulation study and empirical application to S&P 500 data. And lastly, Section 6 concludes with a discussion of open problems, including two that remain unresolved to our knowledge in the existing literature.

2 Background and Framework

The central research questions addressed by Wu and Li (2024) are the following.

1. Does Gaussian optimality, established by Wang *et al.* (2020) for LQ problems under

full information, survive when the state space is augmented by a filtered belief state and the value function is no longer analytically tractable?

2. Can the martingale-based actor–critic architecture of Wang and Zhou (2020) be transferred to the partial-information setting without modification to its core update rules?

3. Does accounting for regime uncertainty via Wonham filtering provide material out-of-sample performance improvements over a single-regime RL baseline?

The model, methodology, and structural results below are organized around these questions.

2.1 Exploratory Stochastic Control

Wang *et al.* (2020) introduce the exploratory stochastic control framework to formalize exploration in continuous-time RL. The control is extended from a point action $u_t \in U$ to a distribution-valued policy $\pi_t \in \mathcal{P}(U)$, and the state dynamics are redefined so that drift and diffusion coefficients are averaged against π_t. For a controlled diffusion with state $X_t \in \mathbb{R}^d$, the exploratory dynamics take the form

$$dX_t = \int_U b(X_t, u)\, \pi_t(du)\, dt + \int_U \sigma(X_t, u)\, \pi_t(du)\, dW_t, \tag{2.1}$$

where b and σ are the drift and diffusion coefficients of the original system. Notably, the exploratory SDE introduces two independent noise sources: one inherited from the environment and one from the randomization of actions, which together produce the Gaussian optimality result downstream.

A temperature-weighted Shannon entropy bonus,

$$H(\pi_t) = -\int_U \pi_t(u) \ln \pi_t(u)\, du, \tag{2.2}$$

is added to the running reward with weight $\Lambda > 0$, yielding a regularized objective that penalizes premature collapse to a deterministic policy. The temperature Λ controls the exploration–exploitation tradeoff: larger Λ sustains stochastic policies longer, while $\Lambda \to 0$ recovers the deterministic optimal control. A key result of Wang *et al.* (2020) is that, for linear-quadratic (LQ) problems, the optimal exploratory policy is Gaussian with analytically tractable mean and variance, and the value function satisfies a modified HJB equation incorporating the entropy regularization. This Gaussian optimality result is the structural fact that makes the downstream algorithms of Wang and Zhou (2020) and Wu and Li (2024) computationally tractable.

2.2 Mean-Variance Portfolio Selection under Full Information

Wang and Zhou (2020) apply the exploratory framework to the continuous-time mean-variance (MV) portfolio problem, producing the Exploratory Mean-Variance (EMV) algorithm. The classical static MV problem of Markowitz (1952) seeks a portfolio weight

vector $w \in \mathbb{R}^n$ minimizing variance $w^\top \Sigma w$ subject to a target expected return $m^\top w \geq \mu_b$ and full investment $\mathbf{1}^\top w = 1$. In the continuous-time setting, the investor's wealth X_t evolves as a controlled SDE driven by risky asset price processes, and the goal is to select a portfolio policy minimizing the terminal mean-variance criterion $\mathrm{Var}(X_T) - \omega \mathbb{E}[X_T]$ for a risk-aversion parameter ω.

Under the exploratory framework, the optimal policy is Gaussian and the value function admits a semi-analytical form. Wang and Zhou (2020) show that the optimal value function satisfies

$$V^*(t,x) = (x-\omega)^2 \frac{1}{f(t)} + g(t) - (\omega - z)^2, \tag{2.3}$$

where $f(t)$ and $g(t)$ are scalar functions satisfying ODEs with known closed-form solutions when market parameters are observed. Because parameters are typically unknown in practice, Wang and Zhou (2020) propose learning f and g through actor–critic updates driven by the martingale characterization of Jia and Zhou (2022): Under any fixed policy π, the process $V^\pi(t, X_t) + \int_0^t r^\pi(s, X_s)\, ds$ is a martingale, and residual drift along observed trajectories signals a mis-specified value function. The EMV algorithm minimizes this martingale loss to evaluate the current policy and applies a score-function gradient step to improve it. The martingale loss serves the same role as the HJB residual in Doya (2000), but avoids the need to estimate second derivatives of the value function from data. A critical assumption of EMV is stationarity of market parameters: the single-regime structure means that $f(t)$ and $g(t)$ do not depend on any latent state, which is precisely the assumption that Wu and Li (2024) relax.

2.3 Regime Switching and Partial Information

Financial markets exhibit pronounced non-stationarity, with periods of differing volatility and expected returns that practitioners often associate with qualitatively distinct regimes such as bull and bear markets. Wu and Li (2024) address this by introducing a continuous-time regime-switching model in which market parameters alternate among a finite set of latent states. In each regime k, asset prices follow a geometric Brownian motion with regime-specific drift $b^{(k)}$ and volatility $\sigma^{(k)}$, and the regime evolves as a continuous-time Markov chain with generator Q. The investor observes only the price path, not the latent regime, so the portfolio weight must be measurable with respect to the filtration $\mathcal{F}_t$ generated by prices alone.

The partial information structure is handled by applying the Wonham filter. The Wonham filter estimates the discrete latent state, regime in our situation, based on a set of observations in the continuous space using the method of predict-observe-update similar to Kalman filters. It also allows for a non-linear methodology in the update phase which is desirable in the difficult world of finance. The Wonham filter processes the observable price path to produce the posterior belief $p_t = \Pr(\text{regime} = \text{bull} \mid \mathcal{F}_t)$. Because the regime is hidden, the investor replaces the true Sharpe ratio $\rho_t \in \{\rho_1, \rho_2\}$ with its filtered

counterpart

$$\hat{\rho}_t = (\rho_1 - \rho_2)\, p_t + \rho_2, \tag{2.4}$$

and defines the innovation process $d\hat{B}_t = dX_t/(\sigma u_t) - \hat{\rho}_t\, dt$, which is a standard Brownian motion with respect to the observable filtration. The wealth dynamics then take the form observable to the investor,

$$dX_t = \sigma u_t\big(\hat{\rho}_t\, dt + d\hat{B}_t\big), \tag{2.5}$$

reducing the partial-information control problem to one driven entirely by observable quantities. The belief state itself evolves as an Itô diffusion given by the Wonham filter of Wonham (1964),

$$dp_t = \big(q_{21} - (q_{12} + q_{21})p_t\big)\, dt + p_t(1 - p_t)(\rho_1 - \rho_2)\, d\hat{B}_t, \tag{2.6}$$

where q_{ij} are the generator entries of the Markov chain. A key structural fact is that $p = 0$ and $p = 1$ are both entrance boundaries for this diffusion: a belief process starting in $(0, 1)$ reaches neither endpoint in finite time, so the state space of p_t is the open interval $(0, 1)$ and no boundary conditions are needed for the PDEs that characterize the value function (see Section 3). By the separation principle of Xiong and Zhou (2007), the original partially observed problem is equivalent to a fully observed control problem in the augmented state (X_t, p_t), converting an unobservable-state problem into an HJB equation in the observable pair (x, p). This reduction is the key structural move that allows the EMV machinery to be extended with minimal modification.

3 Preliminary Results

3.1 Semi-Analytical Form of the Value Function

Combining the exploratory framework of Wang *et al.* (2020) with the augmented state space (t, x, p) from Section 2.3, Wu and Li (2024) derive the regime-switching exploratory mean-variance (RSEMV) problem and establish that its optimal value function takes the form

$$V^*(t, x, p) = (x - \omega)^2 \frac{1}{f(t, p)} + g(t, p) - (\omega - z)^2, \tag{3.1}$$

where $f(t, p)$ and $g(t, p)$ now satisfy a pair of linear parabolic PDEs in (t, p) rather than the scalar ODEs of Wang and Zhou (2020). A critical technical complication is that these PDEs are *degenerate*: the second-order coefficient $(\rho_1 - \rho_2)^2 p^2 (1 - p)^2$ vanishes at $p = 0$ and $p = 1$, so standard uniform parabolicity fails and classical existence-uniqueness theory does not apply directly. Wu and Li (2024) resolve this by establishing a probabilistic comparison principle (their Lemma 1) via an Itô's formula argument on a stopped diffusion, and invoking the existence-uniqueness theory of Krylov (2008) for controlled diffusion processes, establishing in their Theorem 1 that f and g each admit unique solutions in $C^{1,2}$ on compact subintervals of $(0, 1)$, with f bounded between 1 and $e^{\bar{\rho}^2(T-t)}$ and both $|f_p|$ and $|f_{pp}|$ bounded uniformly by constants determined by the environment parameters alone. The absence of boundary conditions is not an oversight but a consequence of the

entrance-boundary classification of $p = 0$ and $p = 1$ for the Wonham filter: A trajectory starting in $(0, 1)$ never reaches either boundary in finite time, so the PDEs are well-posed without them.

The belief-state dependence of f is the key structural departure from EMV. The optimal Gaussian policy has mean

$$\mu^*(t, x, p) = \underbrace{-\frac{x-\omega}{\sigma}\big(\rho_2 + (\rho_1 - \rho_2)p\big)}_{\text{myopic component}} + \underbrace{\frac{x-\omega}{\sigma}(\rho_1 - \rho_2)\, p(1-p)\, \frac{f_p(t,p)}{f(t,p)}}_{\text{hedging demand}}, \tag{3.2}$$

decomposing into a myopic component that reflects only the current filtered Sharpe ratio, and a hedging demand driven by $f_p(t, p)/f(t, p)$ that accounts for drift risk induced by future regime transitions. The hedging demand is largest when $p_t \approx 1/2$ (maximum regime uncertainty), reducing risky asset exposure, and vanishes as beliefs concentrate toward 0 or 1, restoring the myopic term. Crucially, setting $\rho_1 = \rho_2$ collapses the hedging term to zero and reduces μ^* to the single-regime mean of Wang and Zhou (2020), confirming that the hedging demand is an exclusively regime-switching phenomenon with no counterpart in EMV.

3.2 Martingale Characterization of Policy Evaluation

Policy evaluation for the RSEMV problem extends the martingale characterization of Jia and Zhou (2022) used in Wang and Zhou (2020), now applied to the augmented state. Under any fixed policy π, the process

$$M_t^\pi = V^\pi(t, X_t, p_t) - \Lambda \int_0^t \mathcal{H}(\boldsymbol{\pi}(\cdot \mid s, X_s, p_s))\, ds \tag{3.3}$$

is a martingale, where the subtracted term accumulates the entropy bonus earned under π up to time t, and the belief state p_t evolves according to the Wonham filter dynamics. Any nonzero drift in M_t^π along observed trajectories signals a mis-specified value function. The structure of this characterization is identical to that used in Wang and Zhou (2020), which is precisely what enables the actor–critic architecture to transfer from the full-information to the partial-information setting with minimal modification to the update rules themselves.

4 Main Results

4.1 The POEMV Algorithm

Because the PDEs for f and g cannot be solved in closed form when market parameters are unknown, Wu and Li (2024) approximate $\ln f(t, p)$ and $g(t, p)$ by low-order polynomials in p and $(T - t)$, reducing policy evaluation and improvement to estimating a small number of scalar coefficients. This polynomial parametrization directly generalizes the scalar-function parametrization of Wang and Zhou (2020): EMV learns two scalar functions of time, while POEMV learns two functions of both time and the belief state. The

POEMV algorithm alternates between two steps, with the Lagrange multiplier ω updated periodically via the self-correcting rule $\omega \leftarrow \omega - \eta_\omega(X_T - z)$, which drives the mean terminal wealth toward the target z. In the critic step, the martingale loss

(4.1)

$$\mathcal{L}(\theta;\phi) = \frac{1}{2}\,\mathbb{E}\left[\int_0^T \left(h\Big(X_T^{\boldsymbol{\pi}^\phi}\Big) - V^\theta\Big(t,\, X_t^{\boldsymbol{\pi}^\phi},\, p_t\Big) - \Lambda \int_t^T \mathcal{H}\Big(\boldsymbol{\pi}^\phi(\cdot \mid s, X_s^{\boldsymbol{\pi}^\phi}, p_s)\Big)\, ds\right)^2 dt\right]$$

is minimized over critic parameter vector θ using full-episode trajectories, where $h(x) = (x-\omega)^2 - (\omega - z)^2$ is the terminal payoff. In practice this integral is approximated by its discretized analogue over the time grid $\{t_0, \ldots, t_K\}$. In the actor step, a score-function gradient update is applied to the Gaussian policy parameters. The polynomial parametrization keeps the parameter count small, ensuring that score-function gradient estimates remain stable and preserving the computational tractability that distinguishes EMV from general neural-network-based actor–critic methods. Wu and Li (2024) assess approximation quality empirically using finite-difference solutions as benchmarks, finding relative mean absolute errors of 1.03% for $\ln f$ and 0.06% for g at polynomial order $m = 2$, justifying this choice throughout.

4.2 Theoretical Properties

Three structural ingredients support the algorithm's design, though a formal end-to-end convergence result remains open (see Section 6). First, the Wonham filter's optimality ensures p_t is the minimal sufficient statistic for the hidden regime, so no information is lost in the filtering step. Second, the martingale loss is a quadratic criterion whose unique minimizer coincides with the true value function parameters, a property inherited directly from Wang and Zhou (2020). Third, the polynomial approximation of $\ln f(t,p)$ and $g(t,p)$ carries an explicit approximation error bounded in terms of polynomial degree. By restricting the value function class to one whose analytic form is known, made available by the Gaussian optimality result of Wang *et al.* (2020), the composition of the Bellman operator with function approximation remains well-posed, avoiding the instabilities that can arise when richer function classes are used without structural constraints. A further consequence of the exploratory framework is that the optimal Gaussian policy has closed-form variance $\Lambda f(t,p)/(2\sigma^2)$, so exploration decays naturally as $f(t,p)$ decreases toward zero near the terminal time.

5 Simulation Study and Applications

5.1 Simulation Study

Wu and Li (2024) first conduct a controlled simulation study to assess the finite-sample convergence behavior of POEMV before turning to real market data. The true environment parameters are calibrated from 20 years of daily S&P 500 data using a heuristic regime-identification approach, yielding bull-market drift $\mu_1 = 0.25$, bear-market drift

$\mu_2 = -0.73$, volatility $\sigma = 0.22$, and transition rates $\lambda_1 = 0.36$, $\lambda_2 = 2.89$. With these parameters treated as ground truth, the optimal feedback control is obtained by finite-difference solution of the PDEs for f and g and serves as the convergence benchmark.

Two settings are examined. In the first, the true environment parameters are used directly in the discretized Wonham filter; in the second, parameters are estimated from simulated historical data via the heuristic method. The standard EM algorithm for hidden Markov models is also evaluated as an alternative estimator but produces implausibly large transition-rate estimates—exceeding 100 switches per year—due to the low signal-to-noise ratio at daily frequency, so the heuristic approach is adopted throughout. Under both filter settings, POEMV is trained for 200,000 episodes over a 10-year investment horizon with daily rebalancing ($\Delta t = 1/252$). Convergence of the policy parameters ϕ_1 and ϕ_2 toward their theoretical targets $\phi_1^* = (\rho_1 - \rho_2)/\sigma$ and $\phi_2^* = \rho_2/\sigma$ is demonstrated in both cases, and the Lagrange multiplier ω converges reliably to ω^*. Out-of-sample evaluation on 100,000 independent test paths shows that even under estimated parameters, POEMV achieves a terminal wealth variance of 0.170 with the mean target satisfied, compared to the finite-difference optimal variance of 0.067. The dominant source of performance degradation is filtering error from inaccurate parameter estimates rather than deficiencies in the actor–critic learning itself, a finding with direct implications for the open problem of joint parameter estimation and control discussed in Section 6.

5.2 Empirical Setup

Wu and Li (2024) evaluate POEMV on 30 years of daily S&P 500 index data, using the first 20 years as a training period and the final 10 as an out-of-sample test period. The market is modeled with two regimes (bull and bear), and the Wonham filter is initialized with a diffuse prior. To address data scarcity on a single observed trajectory, Wu and Li (2024) supplement the index with 30 constituent stocks whose estimated beta falls between 0.9 and 1.1, treating their paths as proxies for the index under the universal training approach of Wang and Zhou (2020). Dividing these paths into overlapping 10-year blocks yields approximately 75,000 training episodes, and out-of-sample evaluation is conducted on all 122 paths—the index plus the 121 constituent stocks—observed in the held-out period. The comparison baselines are (1) the EMV algorithm in the improved form of Jia and Zhou (2022), which applies the exploratory mean-variance framework of Wang and Zhou (2020) under the single-regime assumption using the martingale-orthogonality actor–critic update, thereby ignoring regime switching, and (2) a classical plug-in control approach that substitutes pre-estimated parameters into the known analytical solution. The EMV baseline is particularly informative because it isolates the contribution of regime-awareness: both EMV and POEMV share the same exploratory SDE framework of Wang *et al.* (2020) and the same martingale-loss actor–critic architecture, differing only in whether the belief state p_t enters the value function and policy.

5.3 Results

POEMV achieves substantially lower out-of-sample variance of terminal wealth than both baselines while satisfying the target mean return constraint. Under daily rebalancing, POEMV's terminal wealth variance is approximately half of EMV and several times smaller than the classical plug-in approach. Under monthly rebalancing the gap widens further, exceeding an order of magnitude relative to the classical method. Both comparisons show greater POEMV resilience around the March 2020 market dislocation, a period of extreme volatility during which the regime-switching model assigns high probability to the bear state and the filter-driven hedging demand reduces risky asset exposure. The quality of the Wonham filter under estimated parameters is directly illustrated in the published paper's Figure 4, which plots the posterior belief p_t against the true simulated regime over a 10-year path; the belief state tracks regime transitions closely, supporting the finding that filtering error is limited even under the heuristic parameter estimates. Wu and Li (2024) further evaluate POEMV under leverage constraints, capping the leverage ratio at $\ell = 2$ and $\ell = 3$ with a reduced wealth target $z = 4$; POEMV continues to outperform EMV on mean, variance, and Sharpe ratio in both cases. These results confirm the regime-awareness introduced in POEMV—and absent from EMV—provides a material improvement in out-of-sample variance control, validating the extension from the single-regime setting of Wang and Zhou (2020) to the partial-information setting of Wu and Li (2024).

6 Conclusion

This review has examined the POEMV algorithm of Wu and Li (2024) as a recent development in continuous-time reinforcement learning for partially observed mean-variance portfolio optimization. The frameworks of Doya (2000), Wang *et al.* (2020), and Wang and Zhou (2020) provide the theoretical scaffolding on which POEMV rests, and a unifying theme across all of them is that continuous time is not merely a limit of discrete time but a setting with its own geometry—the geometry of diffusion operators and the HJB PDE—that can and should be exploited directly. The Gaussian optimality result of Wang *et al.* (2020) and the martingale characterization of Wang and Zhou (2020) are prime examples of this principle, and both are essential to the tractability of POEMV in the partially observed setting.

We identify four concrete open challenges, two of which remain unresolved at the level of formal theory. First, and most fundamental, the convergence of the POEMV algorithm itself is not formally established. Wu and Li (2024) prove structural properties of the value function (Theorem 1) and verify optimality of the Gaussian policy (Theorem 2), but the end-to-end error of the full learning loop, comprising parameter estimation error in the Wonham filter, polynomial approximation error in f and g, and stochastic gradient estimation error in the actor–critic updates, is never jointly analyzed. These error sources compound in a way that, as of yet, does not have a clean decomposition, such that a

rigorous convergence rate result for POEMV remains an open problem.

Second, the separation principle of Xiong and Zhou (2007) that justifies decoupling filtering from control holds specifically for the precommitted MV objective. It is not known whether an analogous separation principle holds for time-consistent equilibrium strategies in the regime-switching partial-information setting. Wu and Li (2024) sidestep this entirely by restricting to precommitted policies, but the precommitted strategy is suboptimal beyond $t = 0$ due to the time inconsistency of the variance term. Extending the framework to time-consistent strategies would require revisiting the separation principle from the outset, since the equilibrium HJB system for time-inconsistent problems under partial information does not obviously admit a filtering-then-control decomposition.

Third, POEMV sequences filtering and control rather than solving them jointly; a fully model-free treatment would require the Wonham filter posterior and value function parameters to evolve together, a coupling that resists the decomposition the current architecture relies on. Fourth, the two-regime structure and single-asset framework of Wu and Li (2024) are deliberate simplifications; incorporating richer latent structure or multi-asset portfolios would require higher-dimensional state and action spaces that interact non-trivially with the filtering and control problems already identified.

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